

Caylia Bonnick

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EDUCATION

Bachelor of Engineering, Electrical Engineering Co-op

09/2021 – 04/2027

McMaster University, Hamilton

- Coursework: Engineering Design, Machine Learning, Communication Networks, Signal Processing

EXPERIENCE

Order Operations Intern

06/2026 – 09/2026

Medtronic, Brampton

- Executed daily order fulfillment workflows and data validation protocols under medical regulatory compliance standards
- Utilized enterprise database systems to manage field inventory distribution pipelines, execute transactional workflows, and enforce strict data-integrity protocols
- Coordinated regional project schedules and tracked delivery timelines alongside the Canada Order Operations Director
- Analyzed operational metrics and core process workflows to isolate system discrepancies, resolve bottlenecks, and implement continuous process improvements

Data Analyst Intern

05/2026 – 06/2026

EcoSchools Canada, Toronto

- Synthesized certification data into dashboards to track program impact across 1,700+ schools
- Developed database structures to accelerate digital certification assessments
- Conducted gap analyses and research to evaluate program effectiveness and growth opportunities

Manufacturing Process Engineer

05/2024 – 12/2024

Procter & Gamble, Belleville

- Lead investigation into root cause of top consumer complaint by testing hypotheses, recommending, and implementing a solution
- Spearheaded the digitization of product dimension and bag size data collection, enhancing product consistency and consumer experience
- Built multi-linear regression models to forecast product attributes, enabling proactive quality control
- Supported logistical process automation by collaborating with project leads, leveraging strong communication and problem-solving skills to define and overcome challenges

Private Tutor

03/2023 – 12/2024

Tutorax, Remote

- Taught a wide range of subjects to students from grades 1 to 12 in both English and French
- Explained key concepts, helped students through class assignments, and worked with parents to design individualized lesson plans

Joe Fresh Sales Associate

10/2019 – 04/2024

Real Canadian Superstore, Whitby

- Ensured all customers received excellent service through courteous salesmanship, and prompt service, maintaining a 100% positive customer and employer feedback score
- Assigned to “Ship from Store” service based on efficiency and ability to thrive under pressure, allowing the opportunity to further develop strong time management skills
- Managed sales, shipments, and returns through a POS software system

PROJECTS

Autonomous Electric Vehicle

04/2026

Associated with McMaster University

- Engineered an autonomous toy car using an STM32 microcontroller, focusing on collision avoidance
- Integrated ultrasonic sensors, a servo motor, and a motor driver to enable obstacle detection and autonomous navigation
- Programmed control algorithms in Python and C++ to process sensor data and adjust the car's movements
- Collaborated with other electrical and computer engineering students, highlighting teamwork, communication and organizational skills in team settings

Digital Engineering Portfolio

03/2026

Personal Project

- Designed a high-performance personal portfolio using HTML5, CSS3, and Asynchronous JavaScript, featuring a custom CLI-inspired boot sequence and 3D Lottie animations
- Engineered a responsive UI using the Intersection Observer API for fluid transitions and seamless cross-platform compatibility
- Managed deployment and technical documentation via Git/GitHub

Microprocessor Systems Project

04/2024

Associated with McMaster University

- Built a device that scans surroundings and uses serial communication to create a 3D rendering of an enclosed space
- Used a ToF Sensor, a Microcontroller, and a PC to collect, send, receive, interpret, and visualize measurement data
- Wrote C++ and Assembly code to receive and transmit data to the PC and a Python code to store received data, using Open3D for data visualization

LEADERSHIP & INVOLVEMENT

Event Director

09/2025 – Present

McMaster Engineers without Borders

- Coordinated logistics and supported outreach, and fundraising events to engage students

Logistics and Marketing Team Member

09/2023 – 09/2025

McMaster Deep Space Analogue Research Expedition (DARE)

- Organized logistics & sponsorships for deep-space analogue research expeditions

CREST Conference Operations Executive Director

11/2022 – 10/2024

McMaster Women in Science and Engineering (WISE)

- Organized 200+ attendee research conference, coordinated panels & sponsorships

TECHNICAL SKILLS

Programming & Analytics

- Python, FastAPI, TypeScript, SQL, R, MATLAB, C/C++, Verilog, VHDL, Assembly

Web Development

- HTML, CSS, JavaScript, Next.js, React

Tools

- GitHub, Visual Studio Code, Netlify, Render, Uvicorn, Quartus II, ORCAD PSpice, Autodesk Inventor

Hardware

- Oscilloscopes, Signal Generators, Integrated Circuits, Raspberry Pi

Spoken Languages

- Fluent in English and French